



MULTI-AGENT URBAN TRAFFIC SIGNAL OPTIMIZATION IN SIMULATED ENVIRONMENTS

Project ID: 25-26J-523

Checklist 1

B.Sc. (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

January 2025

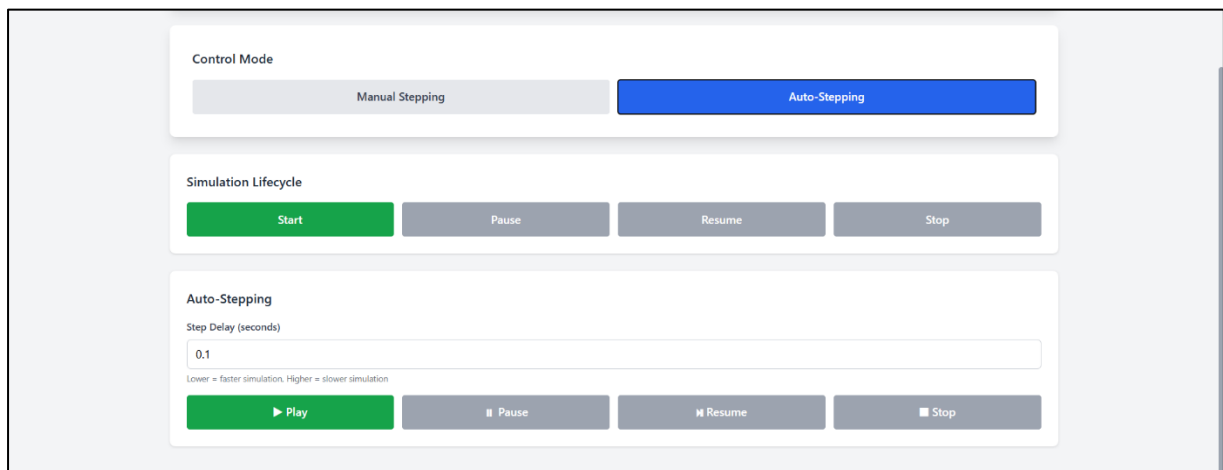
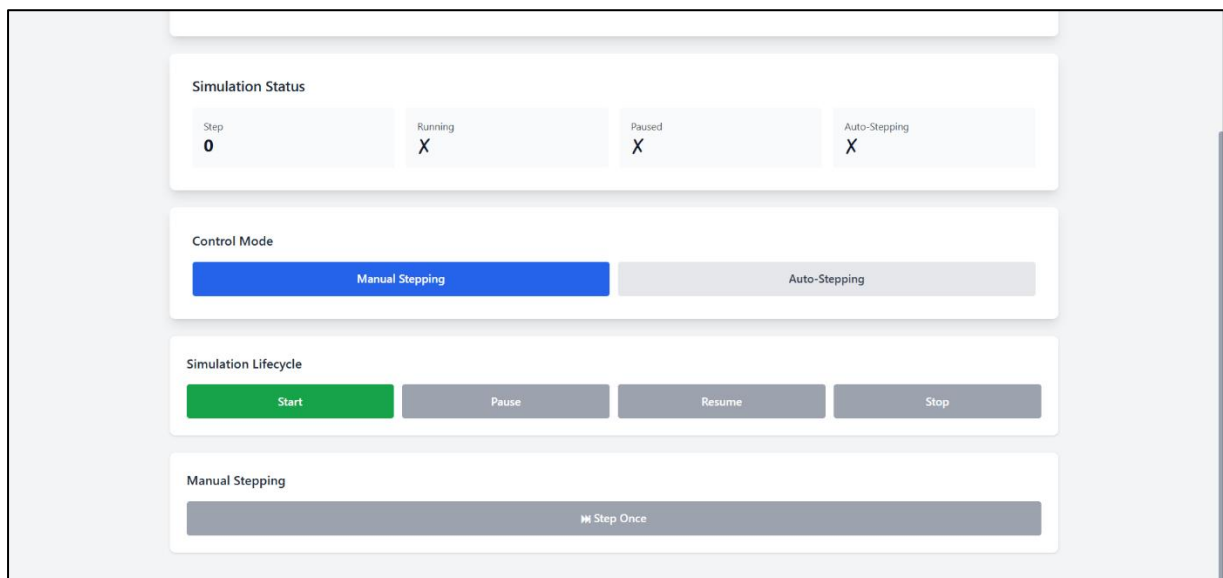
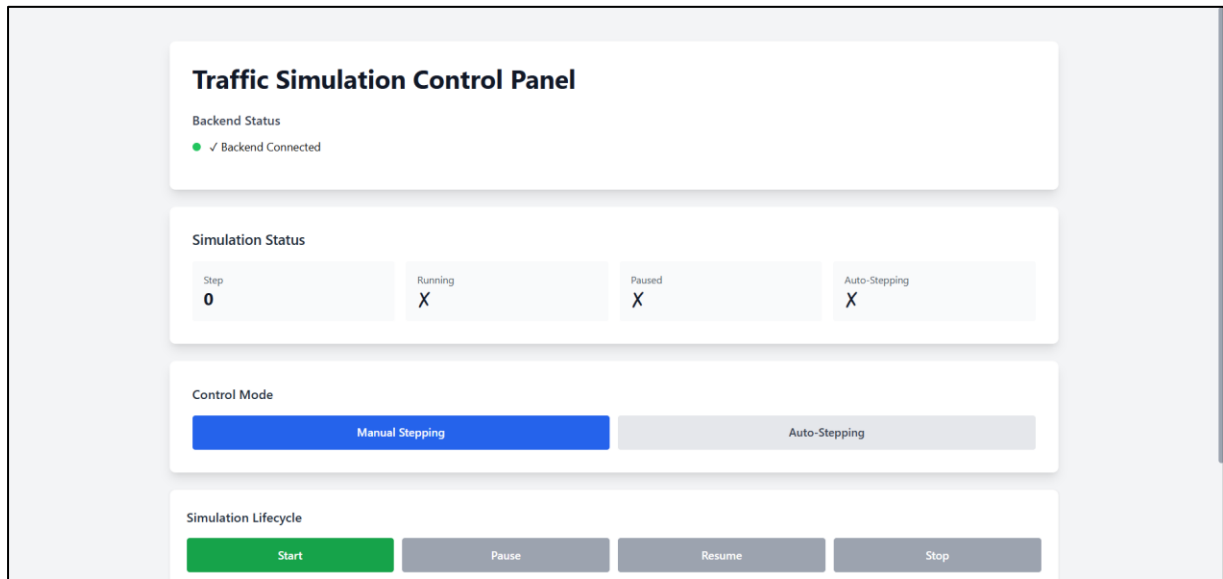
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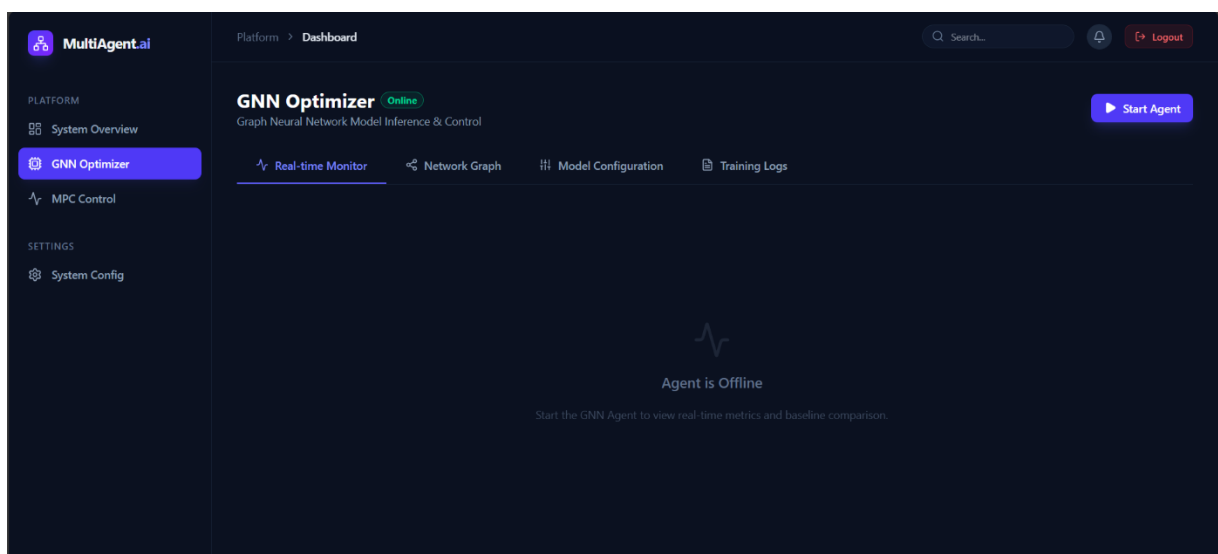
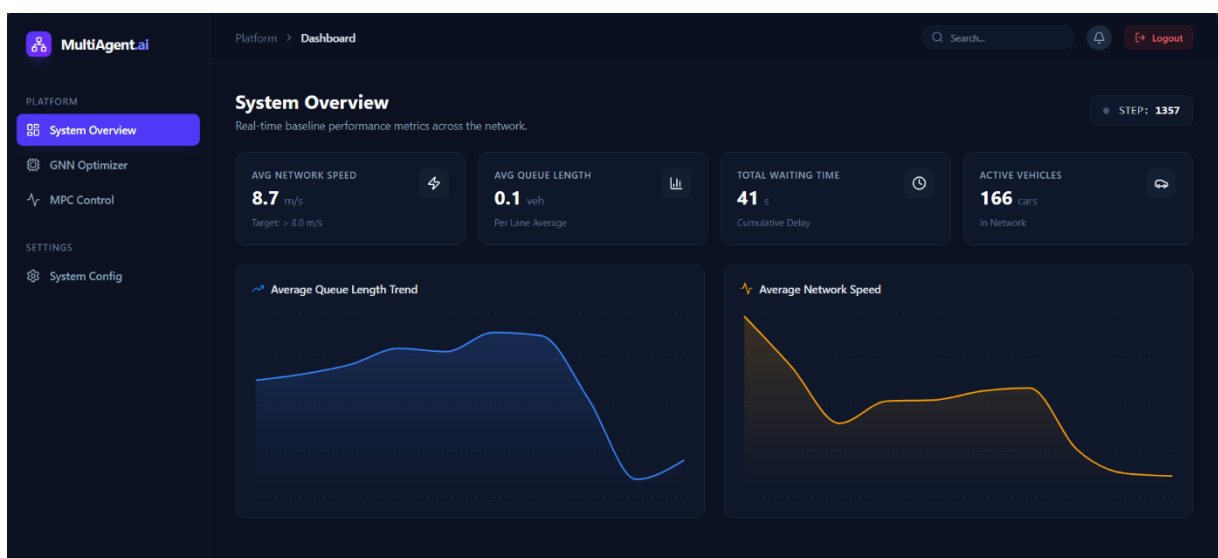
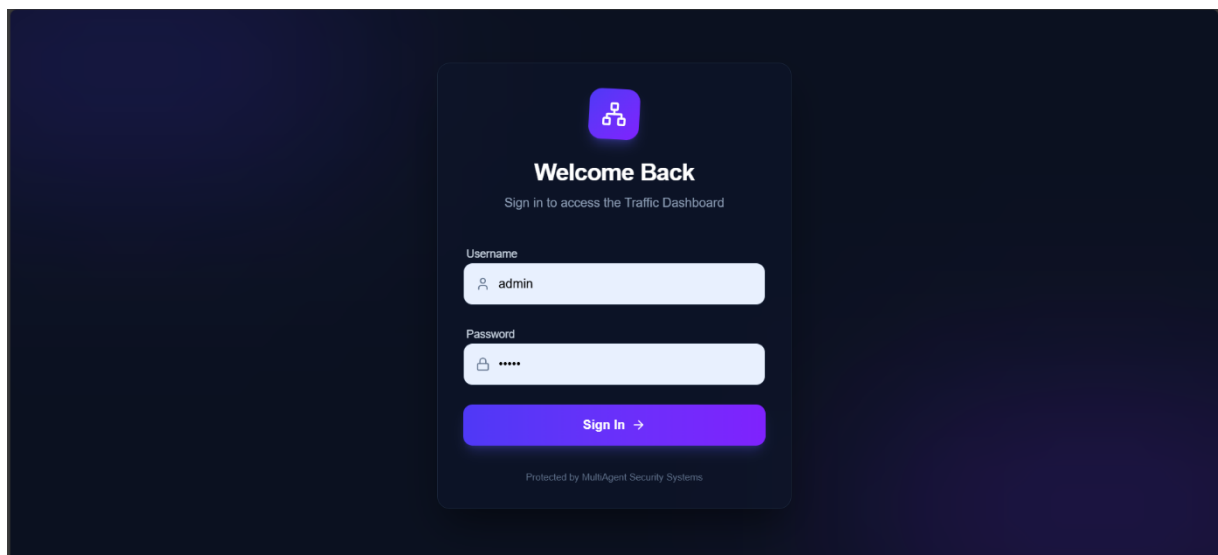
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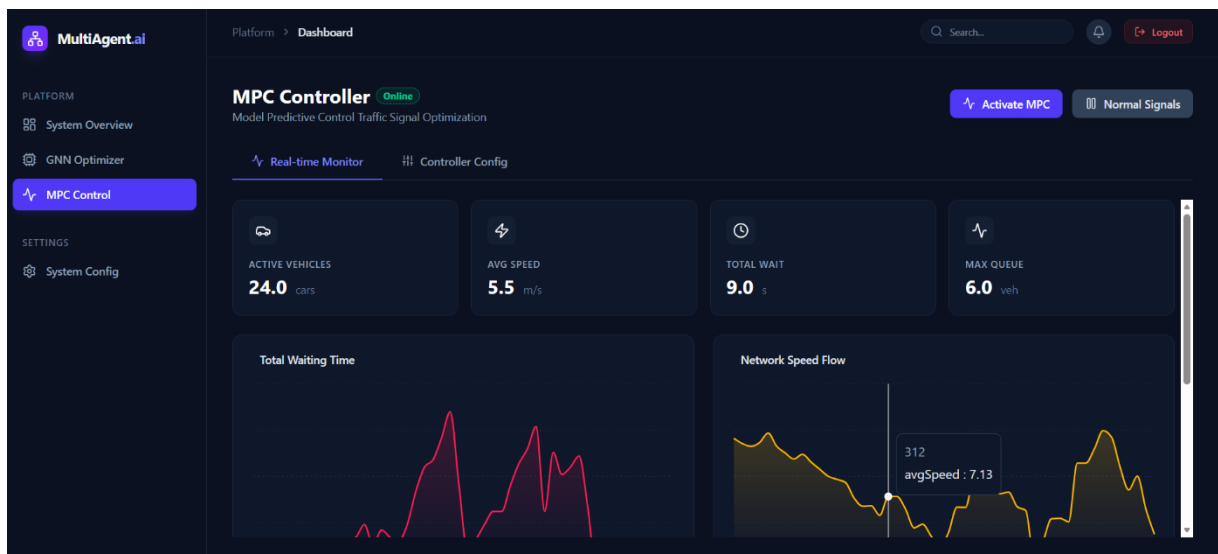
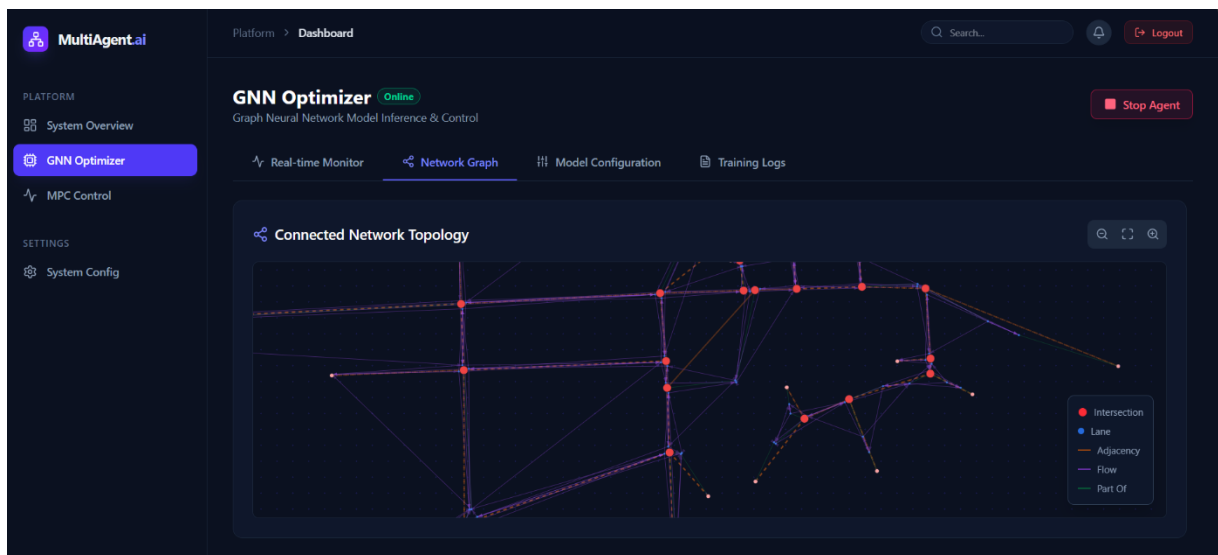
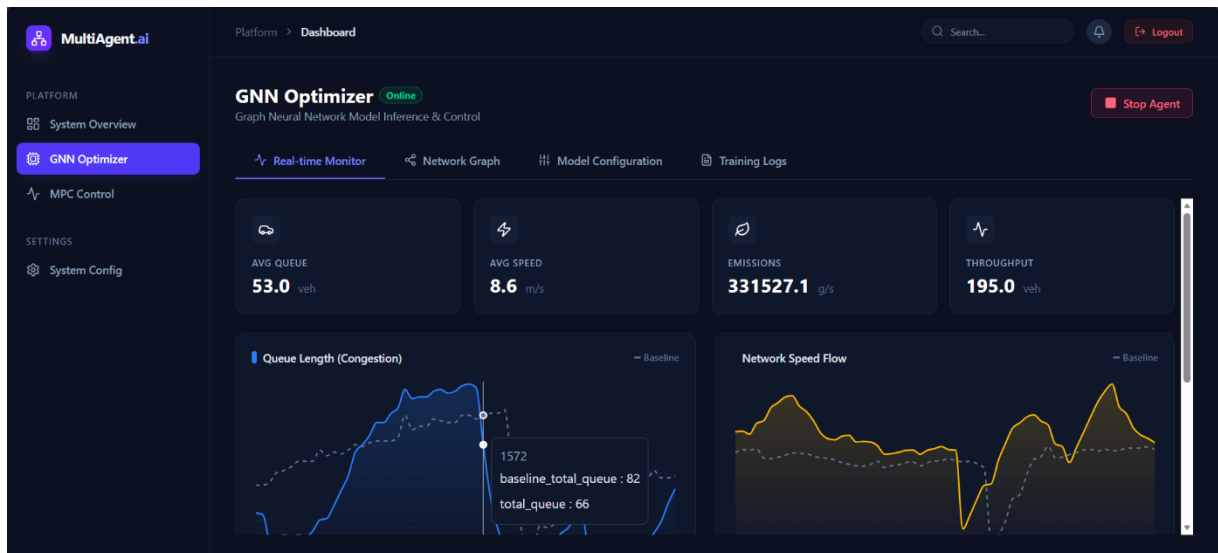
1. Git Repository

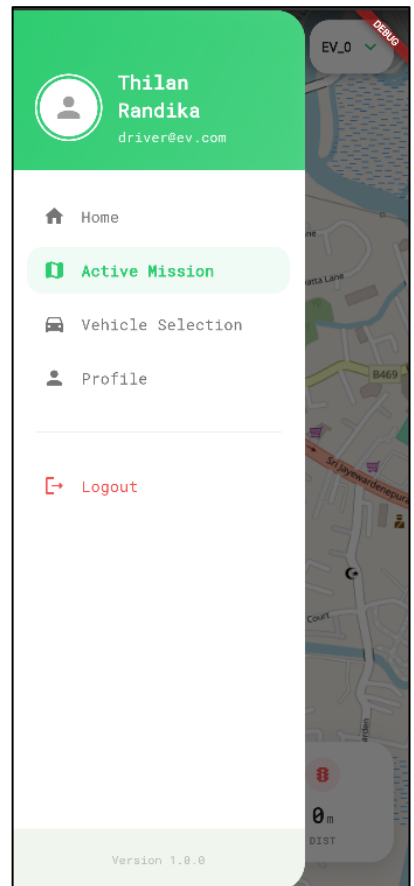
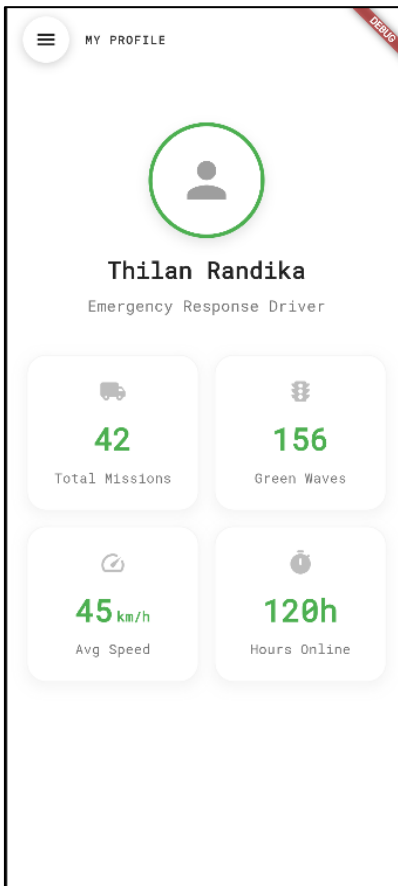
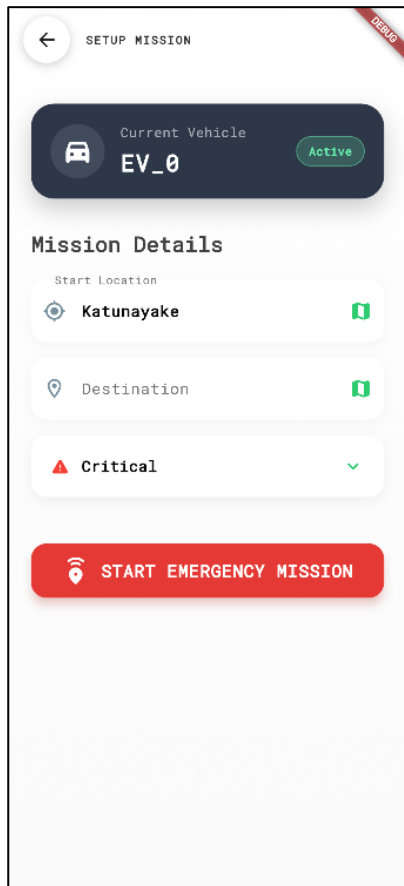
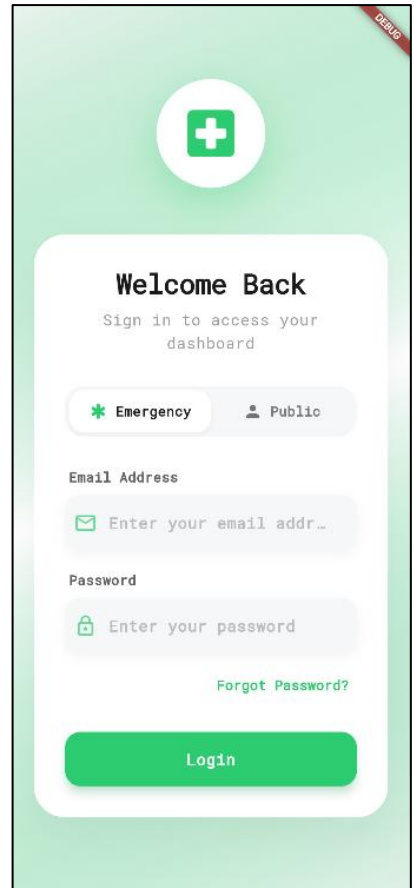
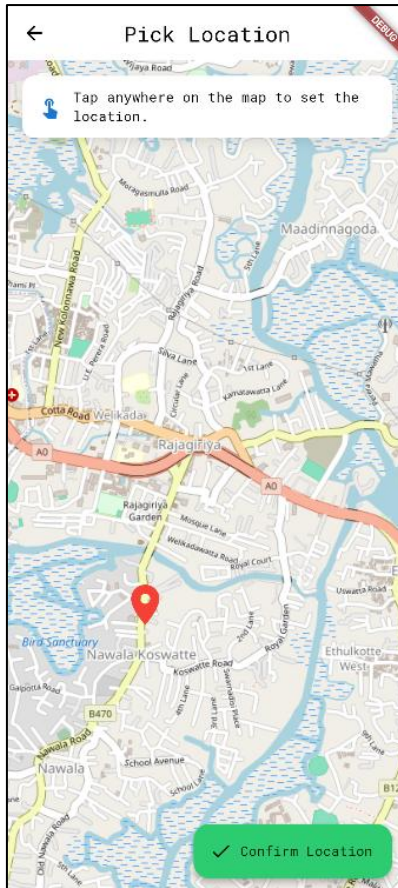
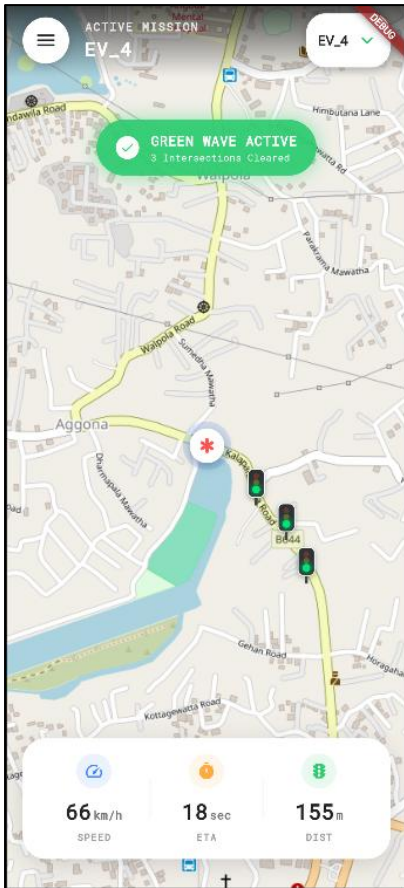
<https://github.com/SLIIT-SE-projects/multi-agent-urban-traffic-signal-optimization-in-simulated-environments.git>

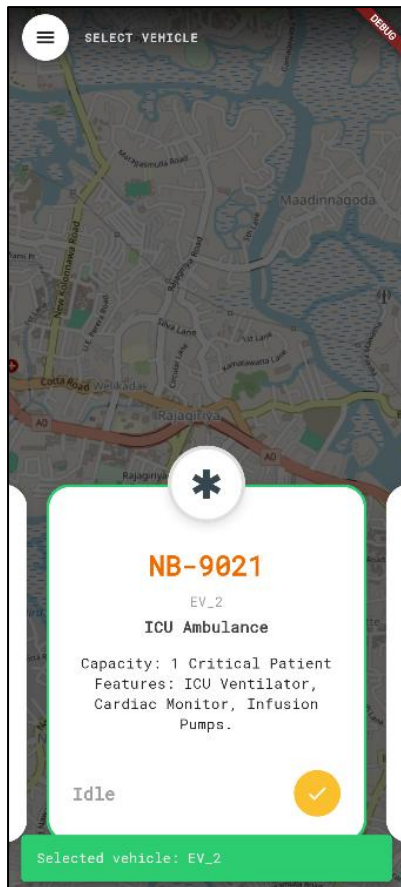
2. Sketches and UI Mockups









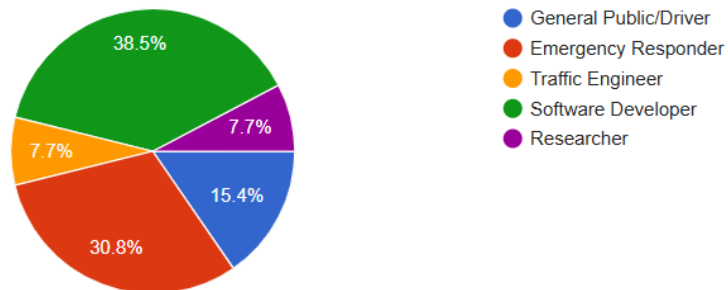


3. User Feedback

Which role best describes you?

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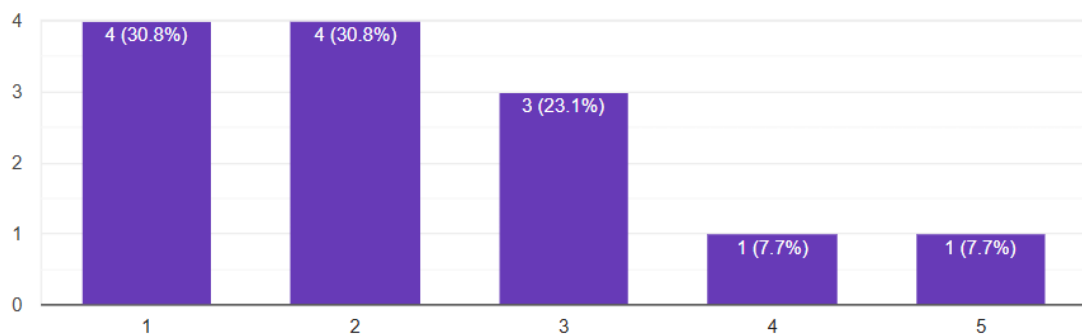
13 responses



How would you rate the efficiency of current traffic signals in your city?

[Copy chart](#)

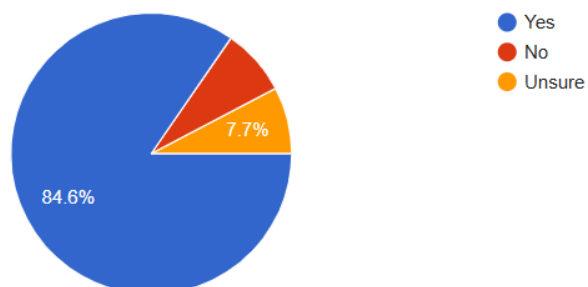
13 responses



Is the use of 'X' to show 'Inactive' status intuitive?

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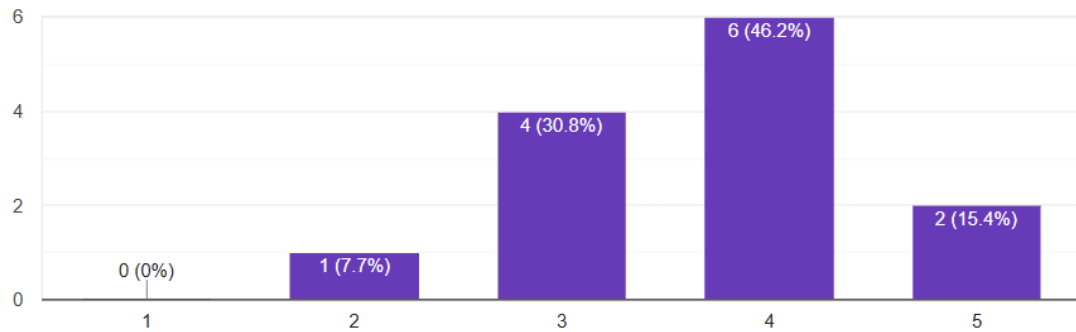
13 responses



How easy is it to identify the active Control Mode (Manual vs Auto) based on the blue highlighting?

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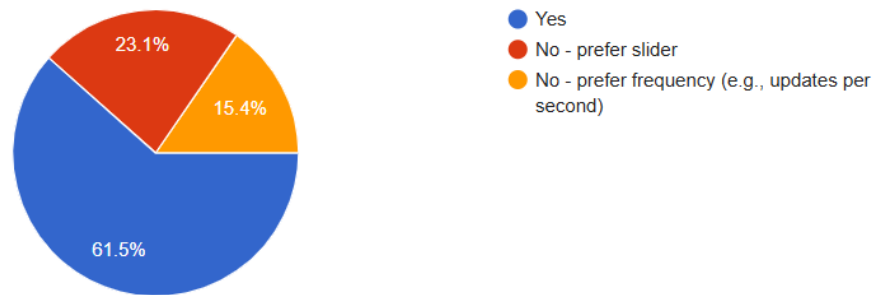
13 responses



Is the 'Step Delay' input intuitive for controlling simulation speed?

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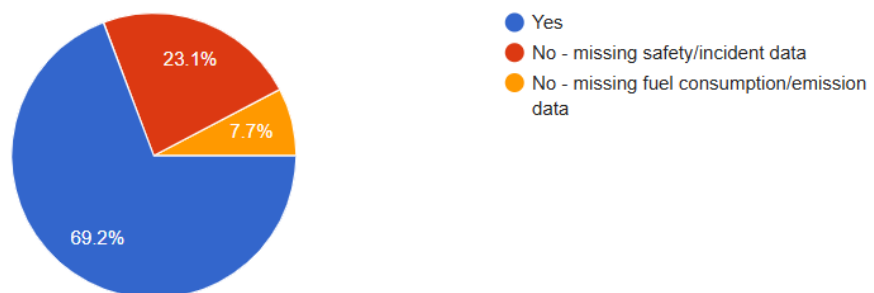
13 responses



Are the key metrics (Speed, Queue, Wait Time) sufficient for a traffic summary?

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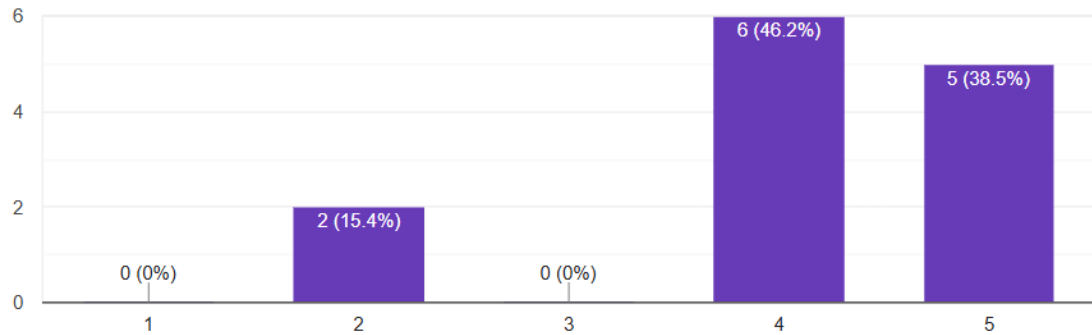
13 responses



How clear is the performance difference between the 'Baseline' (Grey) and 'AI Model' (Blue) lines?

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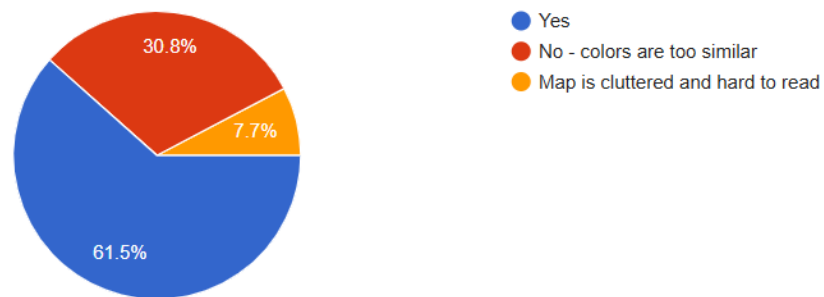
13 responses



Are the colors in the network graph distinct enough to understand?

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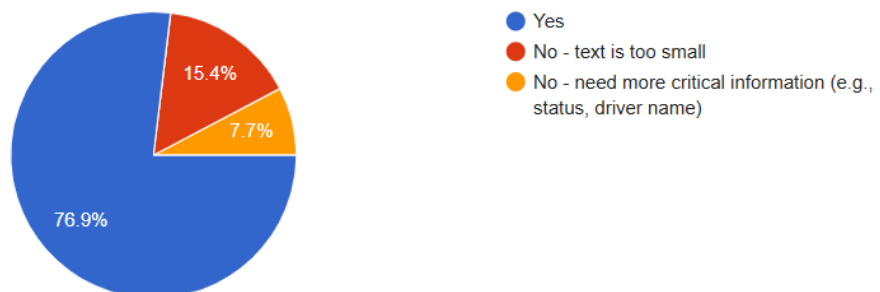
13 responses



Is the vehicle information displayed clearly enough to avoid selecting the wrong ambulance?

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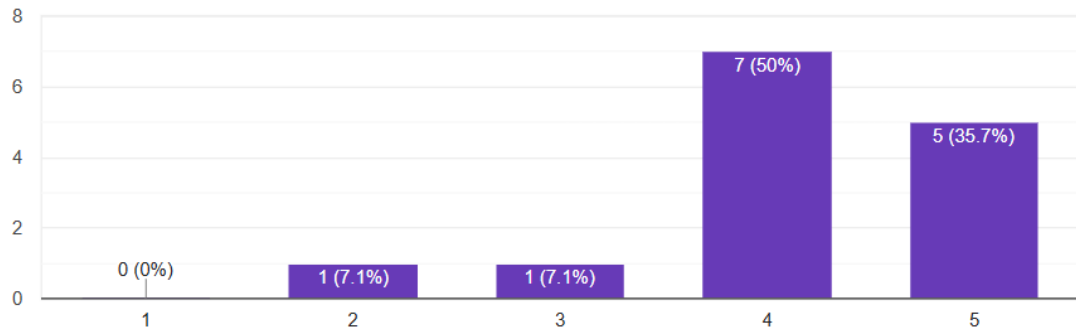
13 responses



Is the 'Green Wave Active' notification banner prominent enough for a moving driver?

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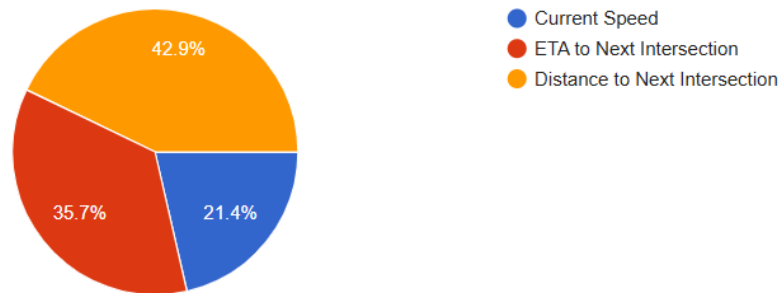
14 responses



Which metric is most critical for you while driving?

 [Copy chart](#)

14 responses



4. Improvements

- **Traffic Simulation and Control Panel:** Developed a web-based interface for real-time traffic simulation monitoring and control parameter adjustments.
- **Traffic Simulation and Control Panel:** Integrated SUMO simulation engine with custom scenarios including grid networks, mega city layouts, and real-world maps.
- **GNN Optimizer:** Implemented Graph Neural Network-based traffic signal optimization with training scripts and worker processes for distributed computation.
- **GNN Optimizer:** Created analysis and comparison tools for evaluating GNN performance against baseline methods.
- **MPC Optimizer:** Built Model Predictive Control system with scenario generation, detector placement, and model training capabilities using SUMO integration.
- **MPC Optimizer:** Developed verification scripts for control logic and performance analysis tools to compare different optimization strategies.
- **Emergency Vehicle Preemption:** Established emergency vehicle priority system with data handling for vehicle tracking and preemption logic.
- **Emergency Vehicle Preemption:** Integrated simulation components for testing preemption scenarios within the traffic control framework.
- **Driver App:** Created cross-platform Flutter application for driver interaction with the traffic system.